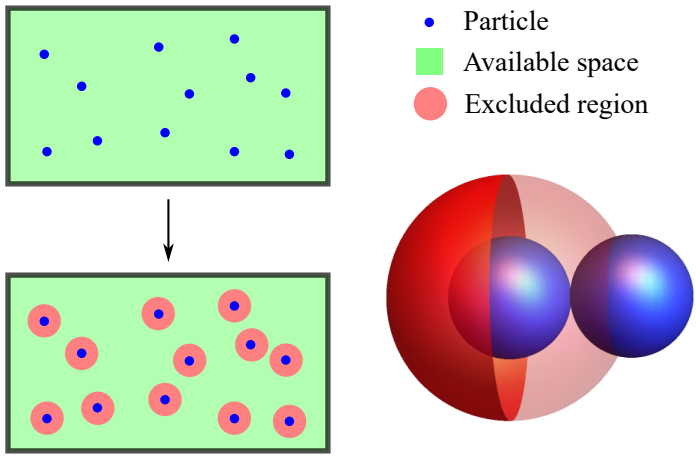
1. Sethna 3.5



In ideal gas model, each gas particles are considered to have no definite volume. This approximation only works when individual particles are separated by sufficiently large distance, that is, when the particles occupy negligible volume. In this question, we do not neglect the space taken up by the gas particles themselves. One way to see how the behavior changes is to compare its equation of state to that of ideal gas.

(a) In order to find the equation of state, we first have to calculate the entropy of the system. Consider a two-dimensional $L \times L$ box with hard walls containing N hard disks of radius $r \ll L$ with summed area $N\pi r^2 \ll L^2$. Let A be the effective area allowed for the disks in the box: $A = (L-2r)^2$, the area allowed for the second disk is $A - \pi(2r)^2$. Observe that each each successive disk has $\pi(2r)^2$ less area than the previous one[1]. Since the particles are identical, we have

$$A_N = \frac{1}{N!}\prod_{n=1}^{N}\left[A - (n-1)\pi(2r)^2\right] ,$$

where the factor $1/N!$ is due to the $N!$ equivalent permutations of placing N disks in a configuration.

(b) The configurational entropy is given by

$$S_{\mathbb{Q}} = k_{\mathrm{B}} \log A_N = k_{\mathrm{B}}\left[\sum_{n=1}^{N} \log\left(A - (n-1)\pi(2r)^2\right) - \log N!\right] .$$

In the dilute limit $N\pi(2r)^2 \ll A$, we have

$$\log\left(A - (n-1)\pi(2r)^2\right) \approx \log A - (n-1)\frac{\pi(2r)^2}{A} ,$$

which gives

$$\sum_{n=1}^{N} \log\left(A - (n-1)\pi(2r)^2\right) \approx N \log A - \frac{N(N-1)}{2}\frac{\pi(2r)^2}{A} .$$

[1] We ignore the case in which the excluded region around the first disk overlaps the excluded region near the walls of the box.

With Stirling's approximation, we find

$$
\begin{aligned}
S_{\mathbb{Q}} &\approx k_{\mathrm{B}}\left(N\log A - \frac{N(N-1)}{2}\frac{\pi(2r)^2}{A} + N - N\log N\right) \\
&= Nk_{\mathrm{B}}\left(1 + \log\frac{A}{N} - \frac{(N-1)}{2}\frac{\pi(2r)^2}{A}\right) \\
&\approx Nk_{\mathrm{B}}\left(1 + \log\frac{A}{N} + \log(1 - \frac{(N-1)}{2}\frac{\pi(2r)^2}{A})\right) \\
&= Nk_{\mathrm{B}}\left(1 + \log(\frac{A}{N} - \frac{(N-1)}{2N}\pi(2r)^2)\right) .
\end{aligned}
$$

where we've recombined the logarithm using the limit $A \gg N\pi(2r)^2$. The effective excluded area is then given by

$$
b = \frac{(N-1)}{2N}\pi(2r)^2 \approx 2\pi r^2 ,
$$

which is half of the excluded area from each disk.

(c) In three dimensions, we have

$$
P = T\frac{\partial S}{\partial V}\bigg|_E .
$$

The analogy of this result in two dimensions is

$$
P = T\frac{\partial S}{\partial A}\bigg|_E .
$$

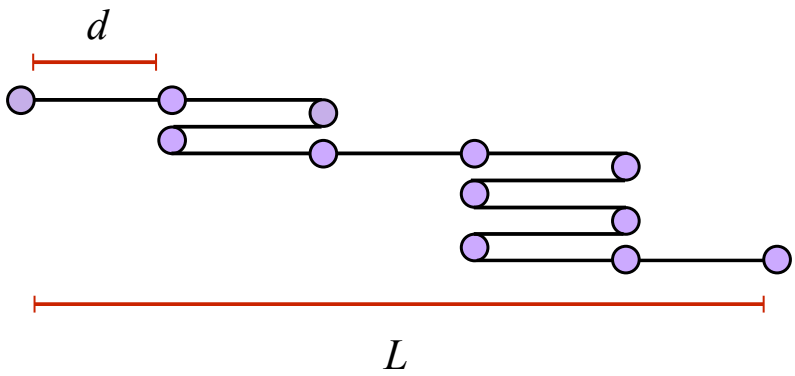
Note that the energy in this system comes purely from kinetic energy $Nk_{\mathrm{B}}T$. The thermal entropy, which arises from the momentum of the particles, therefore do not change as the area changes at constant energy. The change in entropy is then entirely attributed to the change in the number of configurational microstates of the system, which has been calculated in part (a). It is rather straightforward to get

$$
P = T\left(\frac{Nk_{\mathrm{B}}T}{A - bN}\right) \quad \Rightarrow \quad P(A - bN) = Nk_{\mathrm{B}}T .
$$

This reduces to $PA = Nk_{\mathrm{B}}T$ for $b = 0$, which is the familiar equation of state for (two-dimensional) ideal gas. We can further improve this model by taking into account interactions between the particles.

Remarks: The dilute limit $Nd \ll A$ turns out to be crucial for our calculation since we need the gas density to be much lower than that of the liquid state, see Chapter 5.3 in Statistical Physics of Particles (2007) by M. Kardar.

2. Sethna 5.12



Consider a rubber band as a set of N links of length d whereby the link points either parallel or antiparallel to the preceding link with no difference in energy. Let L denotes the extension of the chain, that is, the total change in position to the right from the beginning of the chain to the end.

(a) Suppose n_+ and n_- denote rightward and leftward links, respectively. We have $n_+ + n_- = N$ and $(n_+ - n_-)\, d = L$. The multiplicity in this constant-length ensemble then reads

$$\Omega = \binom{N}{n_+} = \frac{N!}{(N/2 + L/2d)!\,(N/2 - L/2d)!} \ .$$

From $S = k_\mathrm{B} \log \Omega$, we get the configurational entropy:

$$S = k_\mathrm{B} \left[\log N! - \log(N/2 + L/2d)! - \log(N/2 - L/2d)!\right] \ .$$

(b) Note that $dE_\mathrm{bath} = -F dL$. From the definition of inverse temperature:

$$\frac{1}{T} = \frac{\partial S_\mathrm{bath}}{\partial E_\mathrm{bath}} = -\frac{1}{F}\frac{\partial S_\mathrm{bath}}{\partial L} \ .$$

Using the fact that L maximizes the entropy of the universe $\partial S_\mathrm{universe}/\partial L = 0$, we have

$$\frac{\partial S_\mathrm{chain}}{\partial L} = -\frac{\partial S_\mathrm{bath}}{\partial L} \quad \Rightarrow \quad F = T\frac{\partial S_\mathrm{chain}}{\partial L} \ .$$

(c) Applying Stirling's approximation to the configurational entropy in part (a), we find

$$S \approx \frac{k_\mathrm{B}}{2d}\left[Nd \log \frac{4N^2 d^2}{(Nd+L)(Nd-L)} - L \log \frac{Nd+L}{Nd-L}\right] \ .$$

Using the result in part (b), the force F reads

$$F = -\frac{k_\mathrm{B} T}{2d} \log \frac{Nd+L}{Nd-L} \ .$$

In the limit $L \ll Nd$, the equation above simplifies to

$$F \approx -\frac{k_{\mathrm{B}}T}{Nd^2}L = -KL \ .$$

where $K = 1/(\beta N d^2)$. The model turns out to obey Hooke's law in the limit $L \ll Nd$ akin to conventional springs. The actual physical and thermodynamic processes in real rubber bands, however, are more complicated largely owing to its microscopic origins[2].

(d) If T is increased, the chain will contract. As a starting point,

$$\left(\frac{\partial L}{\partial T}\right)_F = -\left(\frac{\partial L}{\partial F}\right)_T \left(\frac{\partial F}{\partial T}\right)_L \ .$$

Observe that

$$\left(\frac{\partial F}{\partial T}\right)_L = \left(\frac{\partial S}{\partial L}\right)_T = -\frac{-KL}{T} \ , \quad \text{where } \left(\frac{\partial L}{\partial F}\right)_T = -\frac{1}{K} \ .$$

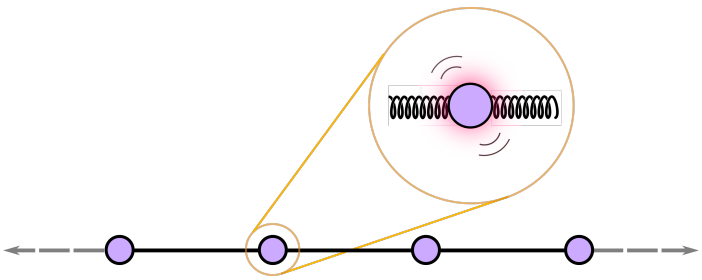
And we find

$$\left(\frac{\partial L}{\partial T}\right)_F = -\frac{L}{T} < 0$$

(e) F,F,T,F,T,T. Note that we are stretching the chain adiabatically, which means there will be no heat flow:

$$S_{\text{total}} = S_{\text{configurational}} + S_{\text{thermal}} \ , \quad \text{where } S_{\text{total}} \text{ is fixed.}$$

As the chain is stretched, there will be less configurations available because the links are getting more constrained. Hence, $S_{\text{configurational}}$ decreases and S_{thermal} increases. Also note that stretching the chain does work on it, and the energy goes into the vibrational modes of the links, which causes T to increase. This is equivalent to the process of compressing an ideal gas adiabatically! Similar argument can also be used to explain the thermodynamic process when the chain contracts[3].



Remarks: In Question 7 and Question 8, we'll solve this problem with canonical ensembles. In Question 6, we'll see that the chain model is similar to the two-state paramagnet.

[2]R. Feynman briefly talked about the physics of rubber bands in Fun to Imagine - 2: Stretching, Pulling and Pushing (1983).
[3]See thermal imaging of a rubber band when it stretches and contracts in Rubber Band Thermodynamics with IR Camera.

3.

Consider a system of N non-interacting molecules in a container of cross-sectional area A. The bottom of the container at $z = 0$ is rigid. The top consists of an airtight piston of mass M which slides without friction.

(a) The Hamiltonian for the system reads

$$H = \sum_{i=1}^{N} \frac{\mathbf{p}_i^2}{2m} + \frac{P_z^2}{2M} + Mgz \ ,$$

where $\mathbf{p}_i$ and P_z denote the momentum of a single gas molecule and the piston, respectively. Here we've neglected the friction and the effect of gravity on gas molecules.

(b) Note that a classical state is a point in phase space (p, q). The single particle partition function for a single gas molecule confined to $V = Az$ is given by

$$Z_1 = \frac{1}{h^3} \int_V d^3q \int_{\mathbb{R}} d^3p \exp\left(-\beta p^2/2m\right) = \frac{Az}{\lambda_{\text{th}}^3} \ ,$$

where $\lambda_{\text{th}} = h/\sqrt{2\pi m k_{\text{B}} T}$. Since the gas molecules are indistinguishable, the partition function of the system reads

$$Z = \frac{1}{N!} \int_{\mathbb{R}} \frac{dP_z}{h} \int_0^\infty dz \, \exp\left(-\beta P_z^2/2M - \beta Mgz\right) \left[\int_{\mathbb{R}} \prod_{j=1}^{N} \frac{d^3p_j}{h^3} \int_V d^3q_j \exp\left(\sum_{i=1}^{N} \mathbf{p}_i^2/2m \right) \right] .$$

Since the term in brackets reduces to Z_1^N, we have

$$Z = \frac{A^N}{N! \lambda_{\text{th}}^{3N}} \left(\int_{\mathbb{R}} \frac{dP_z}{h} \exp\left(-\beta P_z^2/2M\right) \right) \left(\int_0^\infty dz \, z^N \, \exp\left(-\beta Mgz\right) \right) = \frac{A^N}{h \, \lambda_{\text{th}}^{3N} \, (\beta Mg)^{N+1}} \sqrt{\frac{2M\pi}{\beta}} \ .$$

(c) Recall the relation between the Gibbs potential G and chemical potential μ for an ideal gas of N molecules:

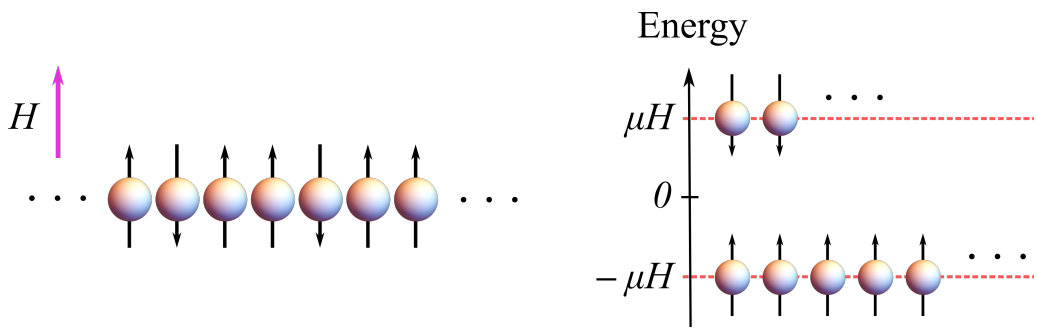
$$G = \mu N, \quad \text{where } \mu = k_{\text{B}} T \log \frac{N \lambda_{\text{th}}^3}{V} \ .$$

While from the result in part (b), we find the thermodynamic potential (in thermodynamic limit)

$$\chi = -k_{\text{B}} T \log Z = -N k_{\text{B}} T \log \frac{A}{\lambda_{\text{th}}^3 \beta Mg} \ .$$

Using the equation of state $PV = N k_{\text{B}} T$, we also find $N/V = \beta Mg/A$. It is now obvious that the potential χ is identical to the Gibbs potential G, i.e. $\chi = G$.

4. Yeomans 2.2



Consider a paramagnetic solid placed in a uniform magnetic field H and contains a large number N of non-interacting, spin-1/2 particles, each of magnetic moment μ on fixed lattice sites. Suppose the field H is pointing in the $+z$ direction, with spin-1/2 particles we then have a two-state paramagnet in which the spin is either "up" ($\uparrow$) or "down" ($\downarrow$).

(a) Note that the field H gives each particle a preference for the state with spin "up". The amount energy required to flip the spin from the "up" state to "down" state is $2\mu H$. For the sake of symmetry, we set the energy $-\mu H$ to the "up" state and μH to the "down" state. The single partition function reads

$$Z_1 = \sum_{\text{states } i} e^{-\beta\epsilon_i} = e^{\beta\mu H} + e^{-\beta\mu H} = 2\cosh(\beta\mu H) \ .$$

We've labeled the particles by assigning a number to each position, the particles are now distinguishable. Since we have an assembly of N particles, the partition function of the solid is simply

$$Z = (Z_1)^N = (2\cosh(\beta\mu H))^N \ .$$

(b) To find the total magnetization, we first need to calculate the average magnetic moment along the direction of H:

$$\langle\mu\rangle = \frac{1}{Z_1}\sum_{\text{states } s} \mu_s e^{-\beta\epsilon_s} = \frac{1}{2\cosh(\beta\mu H)}\left(\mu e^{\beta\mu H} - \mu e^{-\beta\mu H}\right) = \mu\tanh(\beta\mu H) \ .$$

The total magnetization is $M = N\langle\mu\rangle = N\mu\tanh(\beta\mu H)$. The susceptibility χ thus reads

$$\chi = \frac{\partial M}{\partial H} = \beta N\mu^2 \text{sech}^2(\beta\mu H) \ .$$

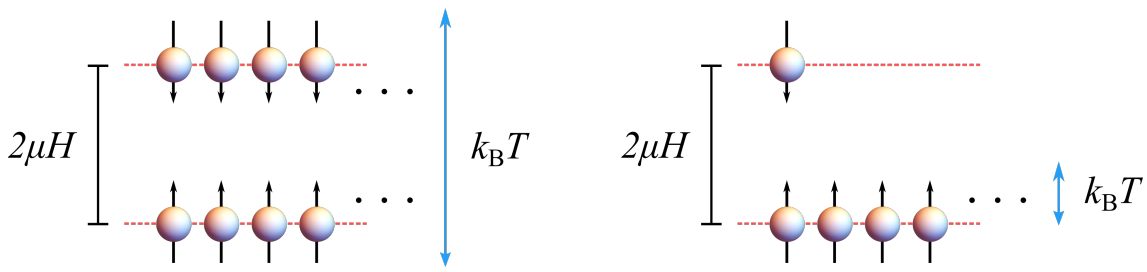
While for the entropy S, we first have to calculate the Helmholtz free energy F for the solid:

$$F = -\frac{1}{\beta}\log Z = -\frac{N}{\beta}\left(\log 2 + \log\cosh(\beta\mu H)\right) \ .$$

It follows

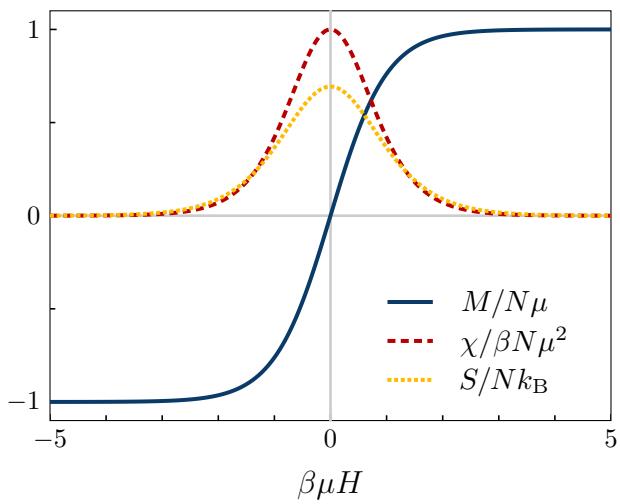
$$S = -\frac{\partial F}{\partial T} = k_{\text{B}}\beta^2\frac{\partial F}{\partial\beta} = Nk_{\text{B}}\Big[\log 2 + \log\cosh(\beta\mu H) - \beta\mu H\tanh(\beta\mu H)\Big] \ .$$

(c) In the limit $\beta\mu H \gg 1$: the total magnetization saturates $M = N\mu$, the susceptibility $\chi = 0$, and $S = 0$. Note that the system is highly ordered in this limit. In the limit $\beta\mu H \ll 1$: the total magnetization follows Curie's law $M = \beta N\mu^2 H$, the susceptibility $\chi = \beta N\mu^2$, and $S = Nk_{\mathrm{B}} \log 2$. In this limit, the two states have equal population and entropy thus maximized.



In the limit $\beta\mu H \gg 1$, excitations triggered by thermal fluctuations will be very rare and the degrees of freedom are frozen out. While for $\beta\mu H \ll 1$, the population of both states are approximately equal. This occurs only if the energy spectrum of the particles is confined within a finite interval, which in our case, equals $2\mu H$.

(d) The total magnetization M, susceptibility χ, and entropy S are plotted below:



All formulas in part (b) make perfect sense for negative T. Rather than viewing temperature as merely a measure of the average energy of the system, we must resort to the general definition

$$\frac{1}{T} = \frac{\partial S}{\partial U} \ .$$

It's easier to draw qualitative physics of this system from its microcanonical ensemble. In particular, we have $U = \mu H(N - 2N_{\uparrow})$ and

$$S = k_{\mathrm{B}} \log \Omega = k_{\mathrm{B}} \log \binom{N}{N_{\uparrow}}$$

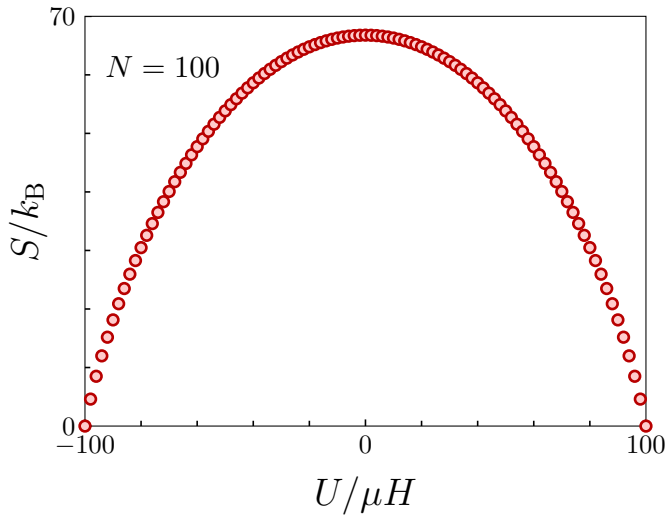


Figure 1: S/k_{B} versus $U/\mu H$ of two-state paramagnet with $N = 100$. The system is stable since $\partial^2 S/\partial U^2 < 0$.

Remarks: The entropy is zero at $U = -N\mu H$ and $U = N\mu H$ when all the particles are in the same state. Observe that the entropy is maximized and symmetric when $N_\uparrow = N/2$ or $N_\uparrow = N_\downarrow$. At $U > 0$, the population of the higher energy level, i.e. $N_\downarrow$, is larger than $N_\uparrow$ (population inversion) and the entropy falls off. From the general definition of T, it approaches ∞ as $U \to 0^-$ and $-\infty$ as $U \to 0^+$. This means the system no longer has the tendency to absorb energy as U approaches zero, and hence $T \to \infty$. In fact, the system tends to give up energy as U increases further. This phenomenon has been observed in a cloud of ultracold quantum gas (see Negative Absolute Temperature for Motional Degrees of Freedom (2013) by S. Braun et. al.) − it cannot take place in natural systems, i.e. systems whose energy has no upper bound, as the system then would need to absorb an infinite amount of energy.

5. Yeomans 2.4

Straightforward calculation.

6. Yeomans 2.6

One of the most important results in the study of the two-dimensional Ising model is that it exhibits a spontaneous magnetization at a finite temperature T_c :

$$\langle s\rangle^8 = 1 - \sinh\left(2J/k_{\mathrm{B}}T\right)^{-4}, \quad \text{for } T < T_c \ .$$

Suppose we denote $K_c = J/k_{\mathrm{B}}T_c$, $t = (T - T_c)/T_c$, and $\zeta = 2K_c$. The above equation can be written as

$$\langle s\rangle^8 = 1 - \sinh^{-4}\left(\frac{\zeta}{1+t}\right)$$

Near T_c, we can expand $\langle s\rangle$ around $t = 0$:

$$\langle s\rangle = \left[1 - \operatorname{cosech}^4(\zeta) - 4\,\zeta\coth(\zeta)\operatorname{cosech}^4(\zeta)\,t - \operatorname{cosech}^4(\zeta)\left(-2\zeta^2 + 10\,\zeta^2\coth^2(\zeta) - 4\,\zeta\coth(\zeta)\right)t^2 + \mathcal{O}(t^3)\right]^{1/8}$$

Note that for Ising model on square lattice, we have the well-known relation for critical temperature:

$$\frac{k_{\mathrm{B}}T_c}{J} = \frac{2}{\log(1+\sqrt{2})} \ .$$

This means $\sinh(\zeta) \approx 1$ and $\coth(\zeta) \approx \sqrt{2}$. The average magnetization $\langle s\rangle$ can then be simplified into

$$\langle s\rangle = \left(-4\sqrt{2}\,\zeta t\right)^{1/8}\left(1 + \frac{18\zeta^2 - 4\sqrt{2}\,\zeta}{4\sqrt{2}\,\zeta}t + \mathcal{O}(t^2)\right)^{1/8} \ .$$

Expanding the terms inside the second bracket around $t = 0$ gives

$$\langle s\rangle = \left(-4\sqrt{2}\,\zeta t\right)^{1/8}\left(1 + \frac{4\sqrt{2}\,\zeta - 18\zeta^2}{32\sqrt{2}\,\zeta}(-t) + \mathcal{O}(t^2)\right) \ .$$

Substituting back $\zeta = 2K_c$, it is now easy to see that $B = (8\sqrt{2}K_c)^{1/8}$ and $b = (1 - 9K_c/\sqrt{2})/8$. The correction term can be ignored when $(-t) \ll |b|^{-1} \approx 0.23$.

Remarks: The value of T_c for two-dimensional Ising model on square lattice can be evaluated from the Kramers-Wannier duality which relates the properties of the model at low and high temperatures. See Chapter 7.4 in Statistical Physics of Fields (2007) by M. Kardar and The Spontaneous Magnetization of a Two-Dimensional Ising Model (1952) by C. N. Yang.